

Experiment 8: Create a Pivot Table and Find Item-wise Units Sold

```
import pandas as pd
```

```
# Sales Data
```

```
data = {
```

```
    'OrderDate':['1-6-18','1-23-18','2-9-18','2-26-18','3-15-18',  
                '4-1-18','4-18-18','5-5-18','5-22-18','6-8-18',  
                '6-25-18','7-12-18','7-29-18','8-15-18','9-1-18',  
                '9-18-18','10-5-18','10-22-18'],
```

```
    'Region':['East','Central','Central','Central','West','East',  
             'Central','Central','West','East','Central','East',  
             'East','East','Central','East','Central','East'],
```

```
    'Manager':['Martha','Hermann','Hermann','Timothy','Timothy',  
              'Martha','Martha','Hermann','Douglas','Martha',  
              'Hermann','Martha','Douglas','Martha','Douglas',  
              'Martha','Hermann','Martha'],
```

```
    'SalesMan':['Alexander','Shelli','Luis','David','Stephen',  
               'Alexander','Steven','Luis','Michael','Alexander',  
               'Sigal','Diana','Karen','Alexander','John',  
               'Alexander','Sigal','Alexander'],
```

```
    'Item':['Television','Home Theater','Television','Cell Phone',  
           'Television','Home Theater','Television','Television',  
           'Television','Home Theater','Television','Home Theater',  
           'Home Theater','Television','Desk','Video Games',  
           'Home Theater','Cell Phone'],
```

```

'Units':[95,50,36,27,56,60,75,90,32,60,90,29,81,35,2,16,28,64],

'Unit_price':[1198,500,1198,225,1198,500,1198,1198,
               1198,500,1198,500,500,1198,125,58.5,500,225],

'Sale_amt':[113810,25000,43128,6075,67088,30000,89850,
            107820,38336,30000,107820,14500,40500,
            41930,250,936,14000,14400]

}

```

```
df = pd.DataFrame(data)
```

```
pivot = pd.pivot_table(
```

```
    df,
```

```
    values='Units',
```

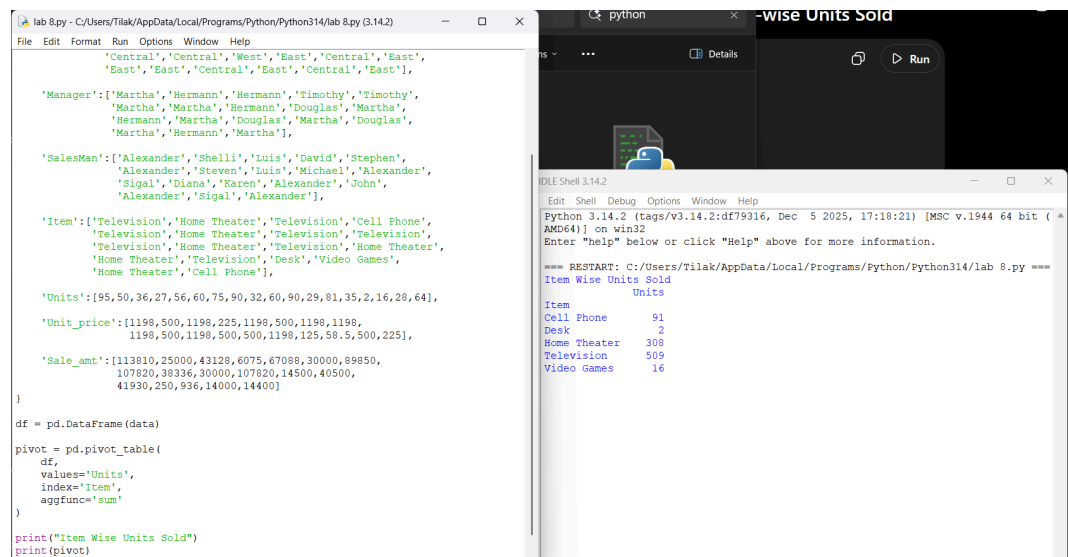
```
    index='Item',
```

```
    aggfunc='sum'
```

```
)
```

```
print("Item Wise Units Sold")
```

```
print(pivot)
```



```

lab 8.py - C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 8.py (3.14.2)
File Edit Format Run Options Window Help

'Central','Central','West','East','Central','East',
'East','East','Central','East','Central','East'],

'Manager':['Martha','Hermann','Hermann','Timothy','Timothy',
'Martha','Martha','Hermann','Douglas','Martha',
'Hermann','Martha','Douglas','Martha','Douglas',
'Martha','Hermann','Martha'],

'SalesMan':['Alexander','Shelli','Luis','David','Stephen',
'Alexander','Steven','Luis','Michael','Alexander',
'Sigal','Diana','Karen','Alexander','John',
'Alexander','Sigal','Alexander'],

'Item':['Television','Home Theater','Television','Cell Phone',
'Television','Home Theater','Television','Television',
'Television','Home Theater','Television','Home Theater',
'Home Theater','Television','Desk','Video Games',
'Home Theater','Cell Phone'],

'Units':[95,50,36,27,56,60,75,90,32,60,90,29,81,35,2,16,28,64],

'Unit_price':[1198,500,1198,225,1198,500,1198,1198,
1198,500,1198,500,500,1198,125,58.5,500,225],

'Sale_amt':[113810,25000,43128,6075,67088,30000,89850,
107820,38336,30000,107820,14500,40500,
41930,250,936,14000,14400]

}

df = pd.DataFrame(data)

pivot = pd.pivot_table(
    df,
    values='Units',
    index='Item',
    aggfunc='sum'
)

print("Item Wise Units Sold")
print(pivot)

```

```

python
python - Item Wise Units Sold
Run

Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

=== RESTART: C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 8.py ===
Item Wise Units Sold
Item      Units
Cell Phone    91
Desk           2
Home Theater  308
Television    509
Video Games   16

```

Experiment 9: Pivot Table

```
import pandas as pd
```

```
df = pd.read_excel(r"C:\Users\Tilak\OneDrive\Documents\sales_data.csv.xlsx")
```

```
print(df.head())
```

```
print(df.columns.tolist())
```

The screenshot shows an Excel spreadsheet with a table of sales data. The table has columns: OrderDate, Region, Manager, SalesMan, Item, Unit_price, Sale_amt, and Units. The data is organized into rows, with some rows grouped by Region and Manager. An IDLE Shell window is open, displaying the execution of the following code:

```
import pandas as pd
df = pd.read_excel(r"C:\Users\Tilak\OneDrive\Documents\sales_data.csv.xlsx")
print(df.head())
print(df.columns.tolist())
```

The output of the code is displayed in the shell window:

```
==== RESTART: C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 9.py ====
      G      H      A      B      C      ...      F
0      OrderDate  Region  Manager  ...  Units  Un
1  2018-06-01 00:00:00  East  Martha  ...    95
2  1198  113810      1-23-18  Central  Hermann  ...    50
3  500  25000      2018-09-02 00:00:00  Central  Hermann  ...    36
4  1198  43128      2-26-18  Central  Timothy  ...    27
5  225  6075

[5 rows x 8 columns]
['A', 'B', 'C', 'D', 'E', 'F', 'G', 'H']
```

Exp-10 Representing Negative values in RED

```
import pandas as pd
import numpy as np
import webbrowser

np.random.seed(10)

df = pd.DataFrame(
    np.random.randint(-50, 51, (10, 4)),
    columns=["A", "B", "C", "D"]
)

def color_negative(val):
    if val < 0:
        return "color:red"
    return "color:black"

styled = df.style.map(color_negative)

styled.to_html("negative_red.html")

webbrowser.open("negative_red.html")

print("HTML file opened in browser.")
```

	A	B	C	D
0	-41	50	-35	14
1	-22	39	43	-21
2	-42	23	-50	-10
3	-14	-34	50	-39
4	4	38	12	-17
5	22	50	28	-1
6	1	4	27	19
7	-37	-25	-37	42
8	36	-20	-20	39
9	-38	15	-19	7

lab 10.py - C:/Users/Tilak/AppData/Local/Programs/Python/Python314/lab 10.py (3.14.2)

```
File Edit Format Run Options Window Help
import pandas as pd
import numpy as np
import webbrowser

np.random.seed(10)

df = pd.DataFrame(
    np.random.randint(-50, 51, (10, 4)),
    columns=["A", "B", "C", "D"]
)

def color_negative(val):
    if val < 0:
        return "color:red"
    return "color:black"

styled = df.style.map(color_negative)

styled.to_html("negative_red.html")

webbrowser.open("negative_red.html")

print("HTML file opened in browser.")
|
```

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